Minimum Spanning Trees

Group 3
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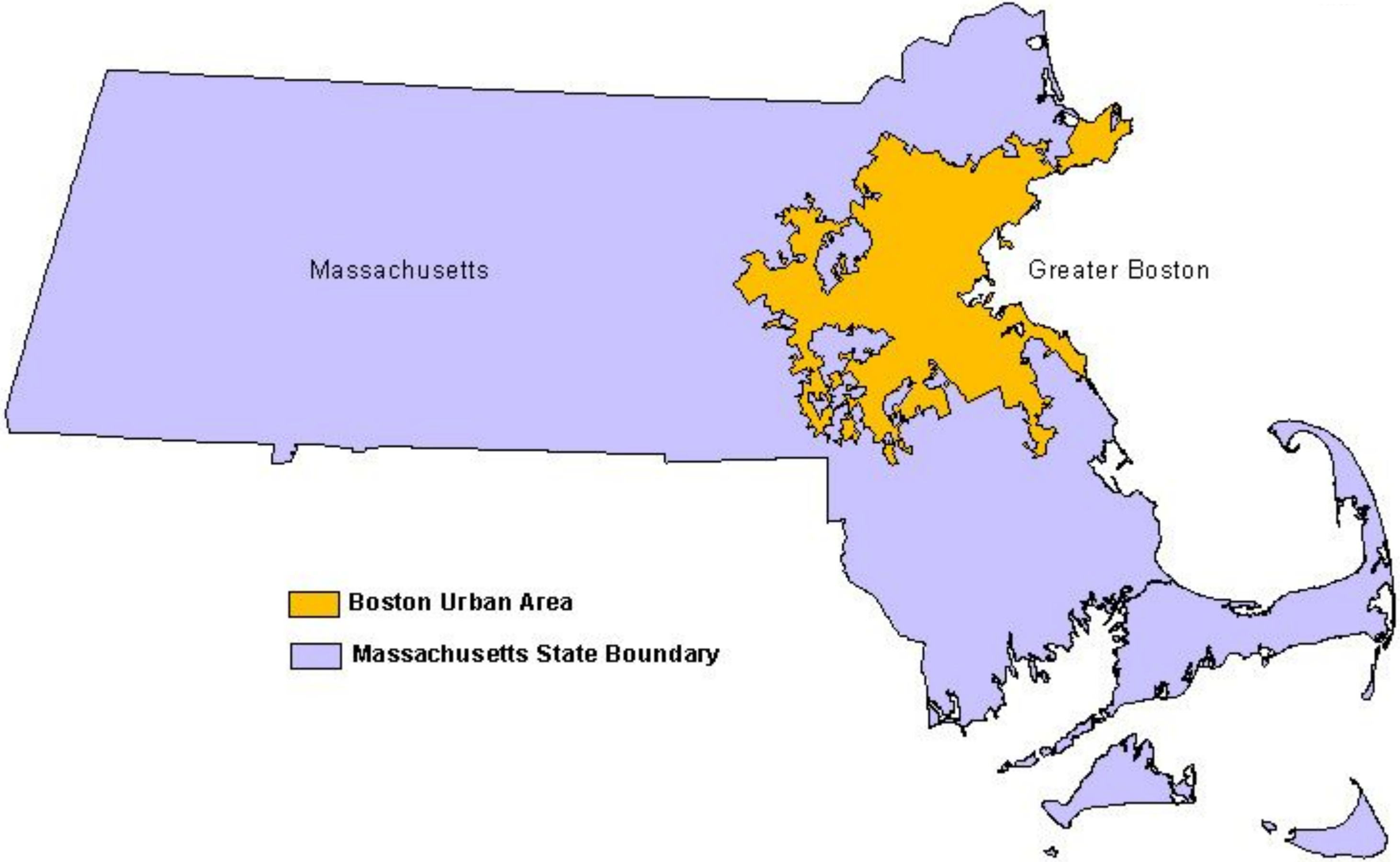
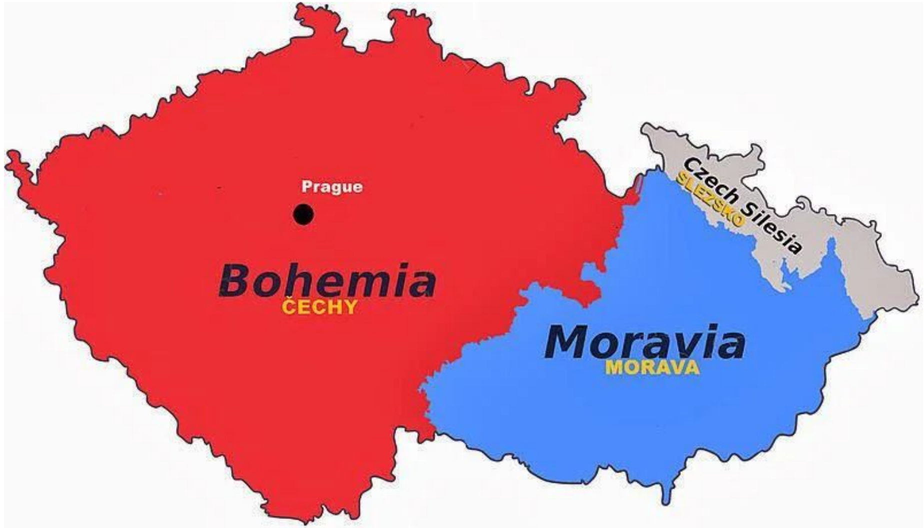


Massachusetts
Greater Boston
Boston Urban Area
Massachusetts State Boundary


Prague
Bohemia
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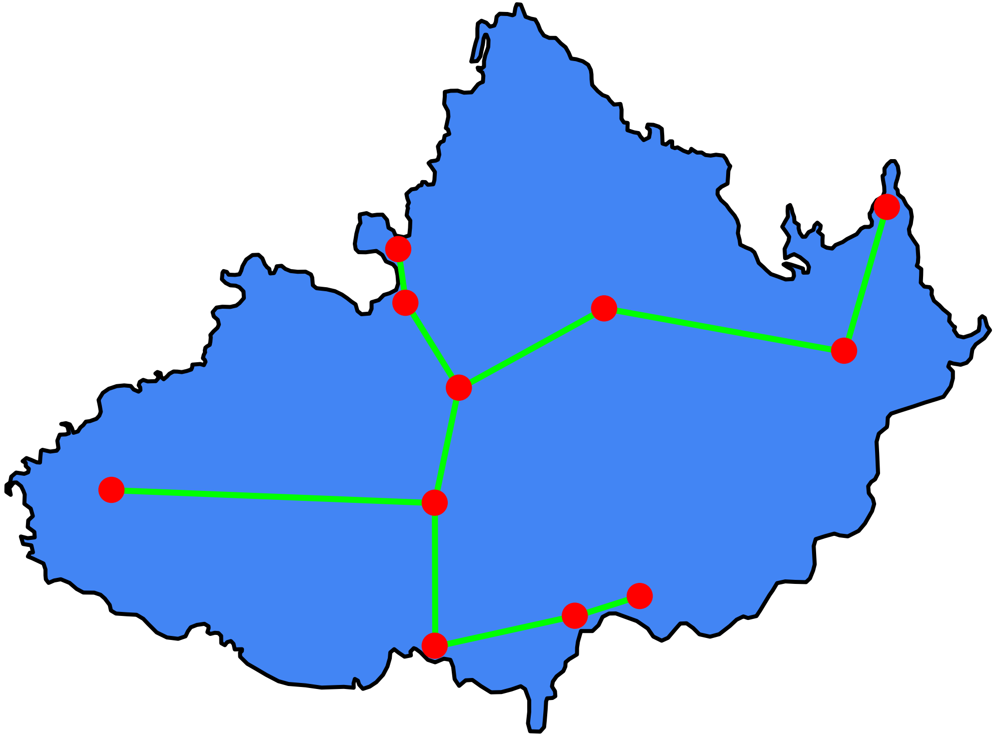
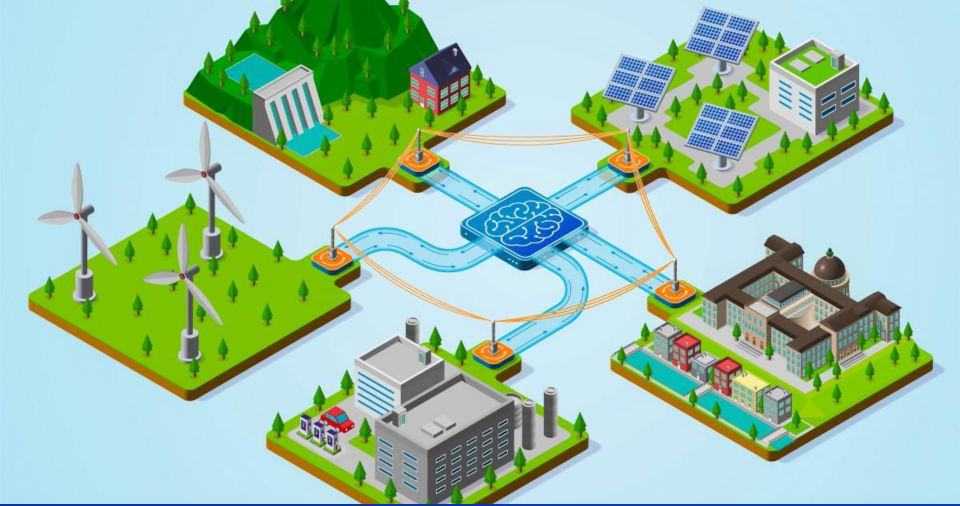
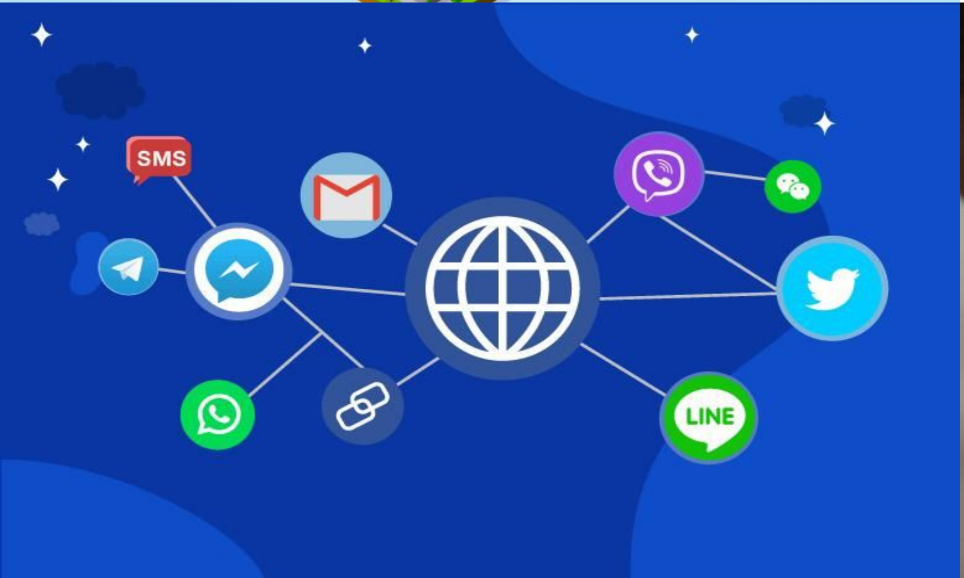
Moravia
MORAVA
Czech Silesia
SLEZSKO

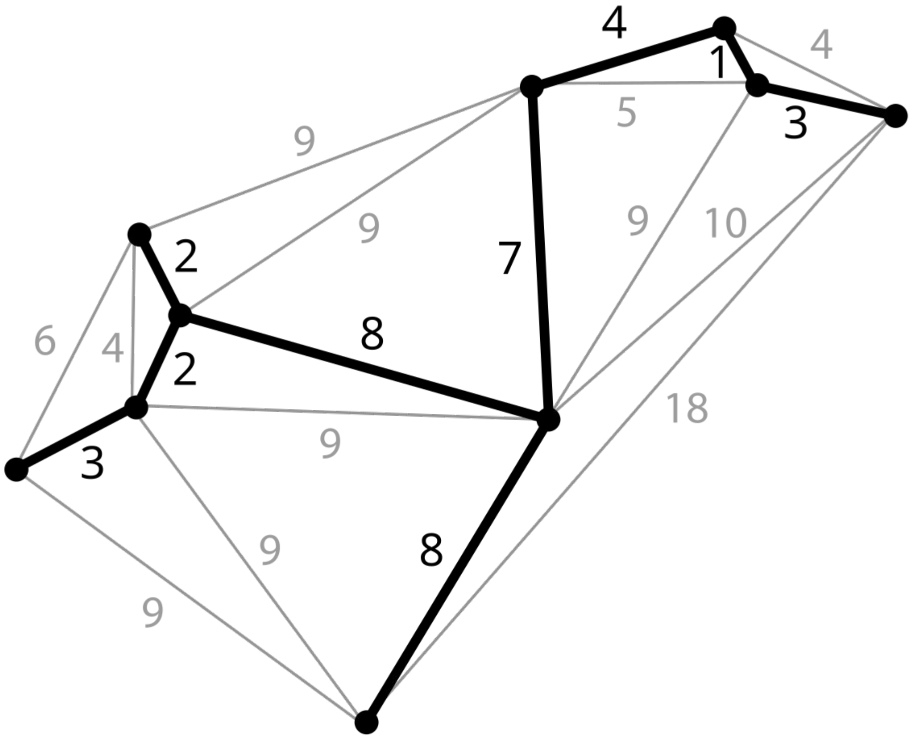




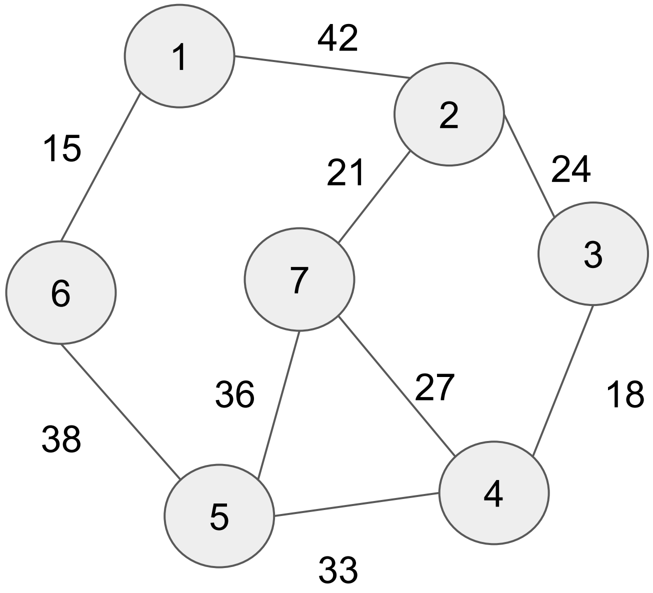

SMS
LINE


ANALIZE
STRUCTURE
ELEMENTS
STATISTIC
REPORT

Minimum Spanning Tree



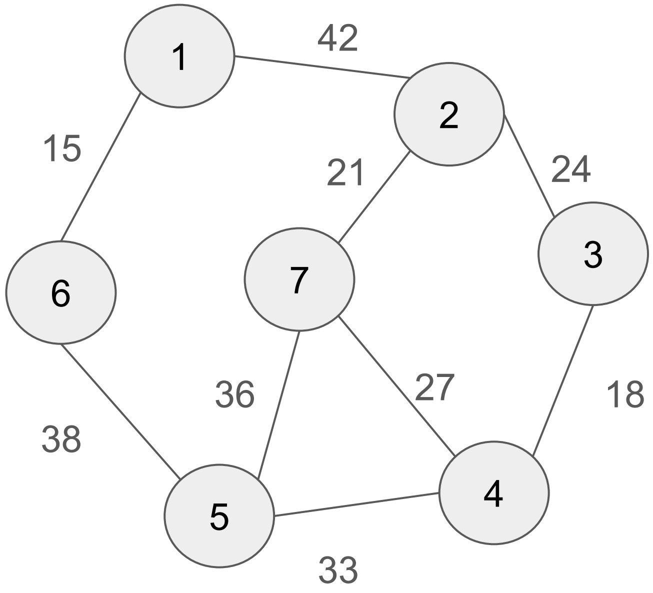
Prim’s Algorithm

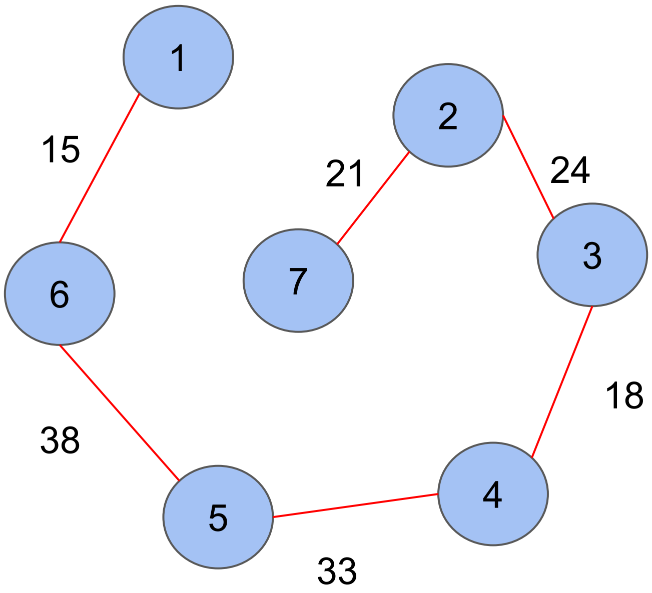


Greedy approach

1. Start with any edge.
2. Add the smallest edge that connects a vertex in the MST to a vertex outside it.
3. Repeat this process until all vertices are included.

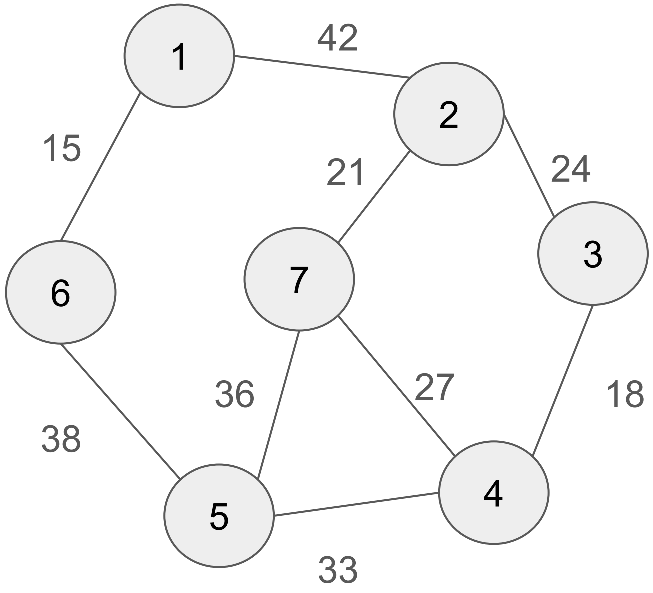
Prim's Algorithm: Example

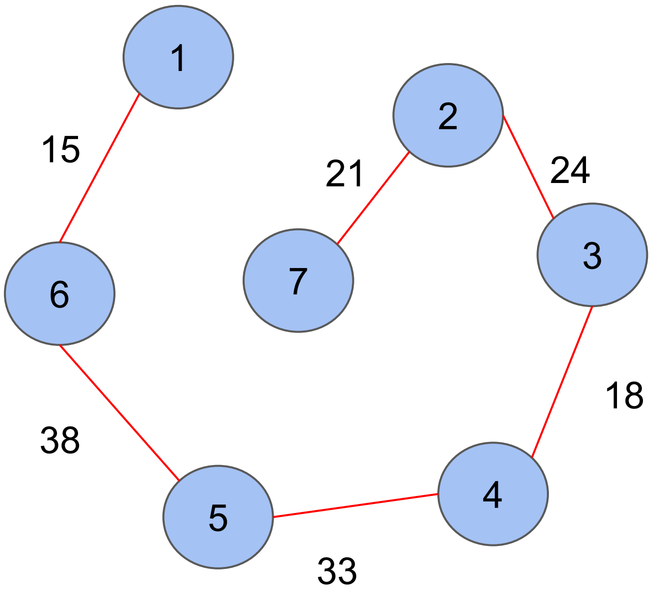




Total Cost = 15 + 38 + 33 + 18 + 24 + 21
= 149

Prim's Algorithm: Example





Total Cost = 15 + 38 + 33 + 18 + 24 + 21
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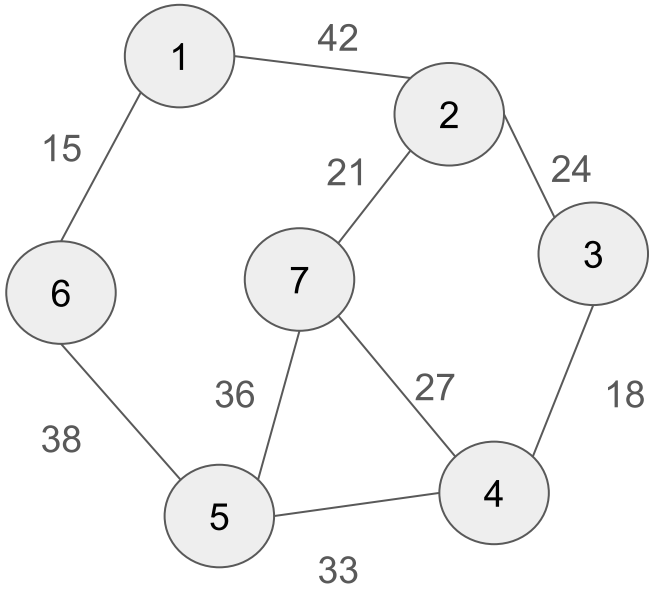
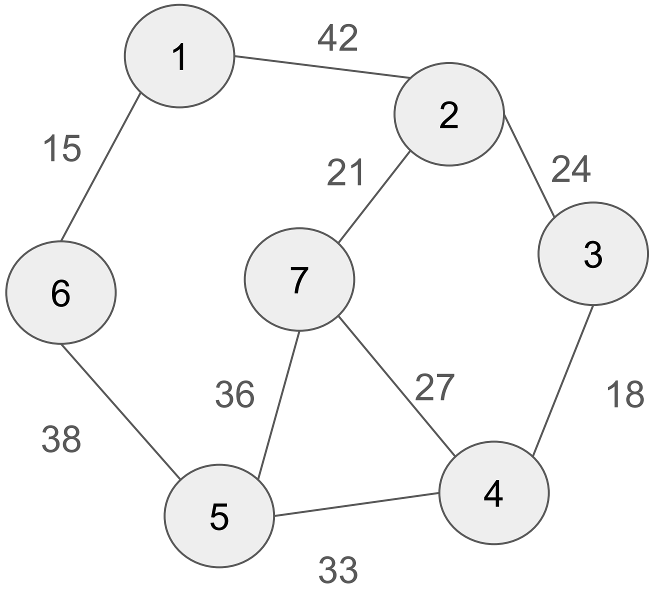
Kruskal’s Algorithm

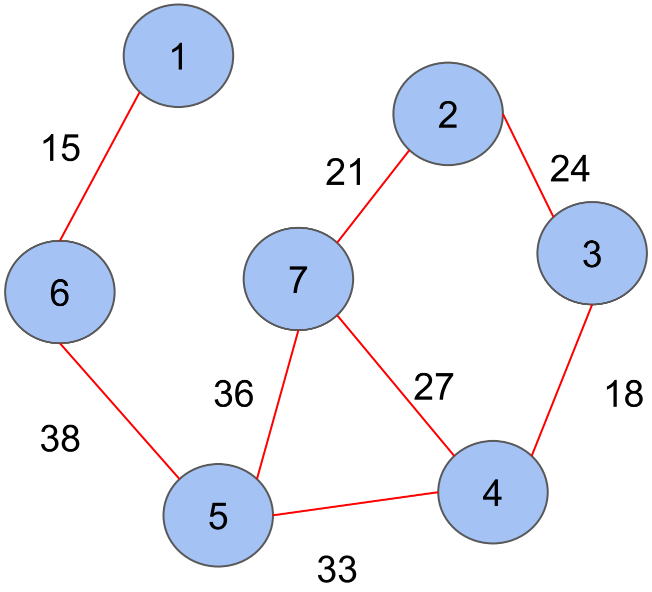


Greedy approach

1. Sort all edges by weight.
2. Add the smallest edge that does not form a cycle with the already selected edges.
3. Repeat this process until all vertices are included.

Kruskal’s Algorithm: Example

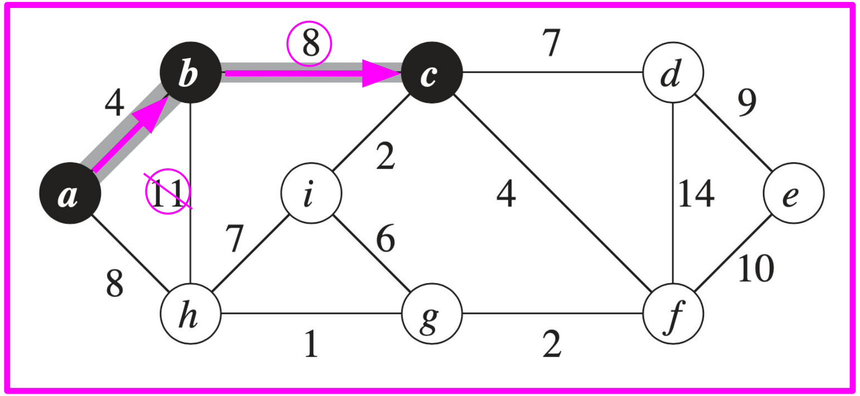




Total Cost = 15 + 38 + 33 + 18 + 24 + 21
= 149

Differences

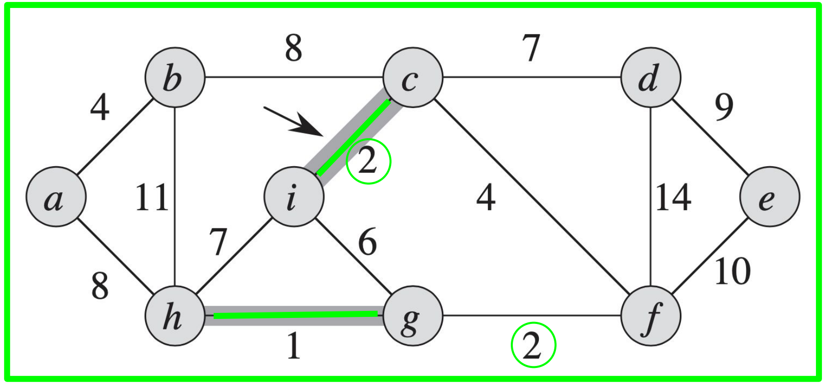
Prim



Start: anywhere

Order: a -> b -> c -> …

Kruskal

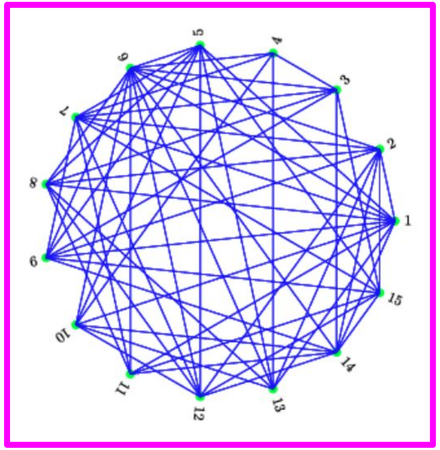


Start: smallest edge

Order: 1, 2, 2, 4…

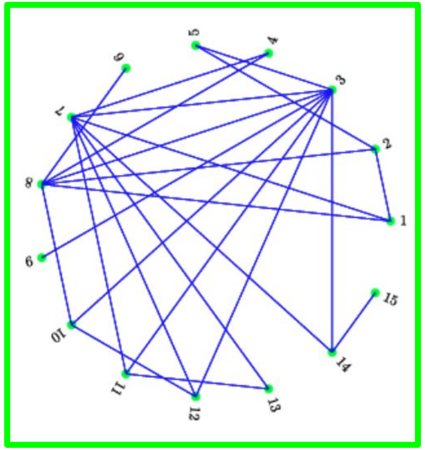
Advantages

Prim



Better for **dense**

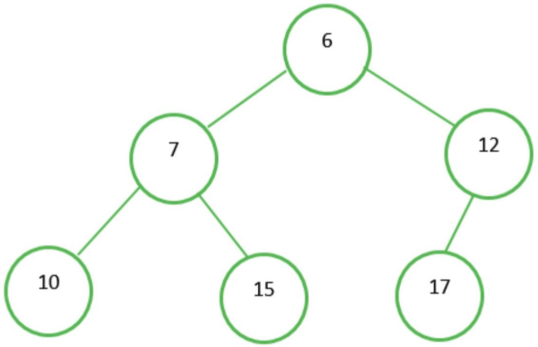
Kruskal



Need to **sort edges** = better for **sparse**

Comparisons

Prim



Priority queue (min-heap)

O(ElogV)

Kruskal

0 1 2 3 4 5 6 7

0—--2 1 3 4 5 6 7 union(0, 2)

0—--2 1 3 4—--6 5 7 union(4, 6)

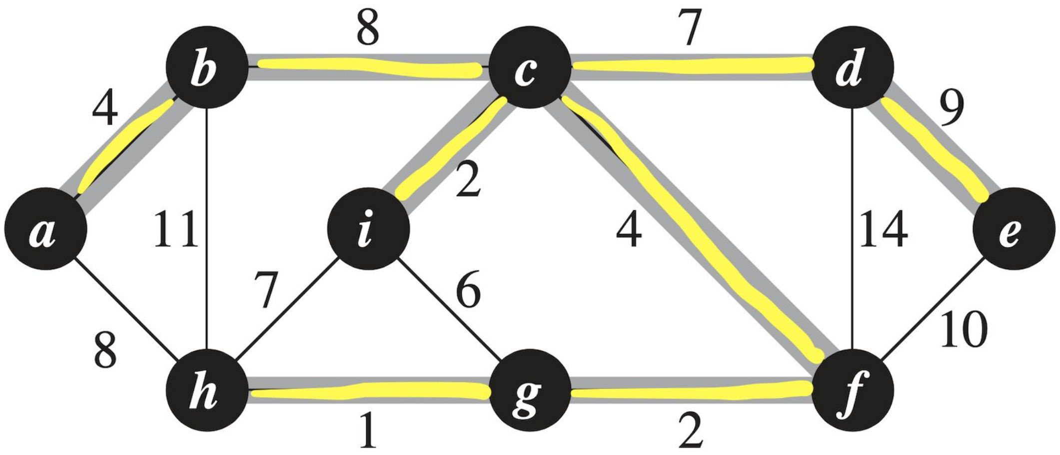
0—--2—-4—-6 1 3 5 7 union(2, 4)

Union-find (disjoint-set union), detects cycles

O(ElogV)

Conclusions of Minimum Spanning Tree

- ***Tree*** connecting ***all vertices*** using ***minimum cost***
- ***Various*** algorithms, ***Prim*** and ***Kruskal***
- ***Greedy***. ***Local*** solution = ***global*** solution



Questions

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